

12	A piece of metal of mass m, specific heat capacity c and temperature 20°C is placed into a liquid of temperature 100°C. The liquid, which is in a well-insulated container, has mass $3m$ and specific heat capacity $2.5c$. What is the temperature of the liquid when thermal equilibrium is reached?				
	A 56°C	B 60°C	C 85°C	D 91°C	
	Answer: D Using $Q = mc\Delta\theta$, heat gained by metal = heat lost by liquid; $mc(\theta_f - 20) = (3m)(2.5c)(100 - \theta_f)$ $\theta_f - 20 = 750 - 7.5\theta_f$ $\theta_f = 91^{\circ}\text{C}$				

13	The graph shows the variation with time t of temperature change $\Delta\theta$ for 1 kg of a substance
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